

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		A, D and F all needed			1	1		
		(ii)		4			1	1		
		(iii)		A		1		1		
		(iv)		H H H H H—C—C—C—C—H H H H H—C—H H accept any correct representation of 2-methylbutane			1	1		
		(v)		add (orange) bromine water / bromine (1) award (1) for any of following - it is decolourised by D only - it goes from orange / brown to colourless in D only - it is decolourised by D but not C - it goes from orange / brown to colourless in D but not C neutral answer – reference to ‘clear’		2		2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			double bond splits / opens / breaks (1) the (vinyl chloride) molecules join / link together / form chains (1) accept 'they join together' $\left[\begin{array}{cc} \text{H} & \text{Cl} \\ & \\ -\text{C} & -\text{C}- \\ & \\ \text{H} & \text{H} \end{array} \right] \quad (1)$ neutral answer – reference to heat / catalyst	3			3		
	(c)			award (1) for any two of following • does not rust / corrode • flexible / easy to shape • lightweight / low density neutral answers weather resistant doesn't need to be painted reference to cost do not accept low melting point doesn't conduct electricity / heat	1			1		
				Question 3 total	4	3	3	10	0	2